Upper-Owner and Optical Cleanup

Analytic ledger replacement, Packet E

Abstract

This note replaces the executable scope checks surrounding the upper owner of the final spherical-shell residual. It proves directly that the eligible upper wall is $K_0(\rho)$ and that $k_{11}(\rho) < K_0(\rho)$ on the required ratio interval. It also records the symbolic identities that make the repeated optical ledger samples unnecessary.

1 The upper owner

For $0 < \rho < 1$, set

$$a_\rho := \frac{2\pi\rho}{1-\rho}, \qquad \eta_\rho := G_1(\max\{\rho, 1/2\}),$$

where

$$G_1(s) = \frac{\sqrt{1-s^2} - s\arccos s}{\pi}, \qquad \omega_0 := G_1(1/2) = \frac{\sqrt{3}}{2\pi} - \frac{1}{6}.$$

Let $C_0 > 0$, and define

$$K_0(\rho) := \left(\frac{\sqrt{a_\rho} + \sqrt{a_\rho + 4\eta_\rho C_0}}{2\eta_\rho}\right)^2. \tag{1}$$

Also put

$$\rho_c := \frac{1}{1+2\pi}, \qquad z_\rho := \frac{\pi}{1-\rho}, \qquad k_{11}(\rho) := \sqrt{z_\rho^2 + 132}.$$

Lemma 1 (The final upper wall is genuinely above the staircase)**.** *For*

$$\rho_c \le \rho < \frac{7}{8},$$

one has

$$k_{11}(\rho) < 64 < U(\rho). \tag{2}$$

More precisely, $U(\rho) = K_0(\rho) > 64$ *for* $\rho_c \le \rho < 5/6$*, while*

$$U(\rho) = \frac{54}{(1-\rho)^2} \ge 1944 \qquad (5/6 \le \rho < 7/8).$$

In particular, on the final surviving interval $\rho_c \le \rho < 39/50$*, the upper owner is* $K_0(\rho)$*.*

Proof. We use only

$$3 < \pi < \frac{22}{7}, \qquad \sqrt{3} < \frac{7}{4}.$$

They imply

$$\omega_0 = \frac{\sqrt{3}}{2\pi} - \frac{1}{6} < \frac{7/4}{6} - \frac{1}{6} = \frac{1}{8}.$$

Since

$$G_1'(s) = -\frac{\arccos s}{\pi} < 0 \quad (0 \le s < 1), \qquad G_1(1) = 0,$$

one has $0 < \eta_\rho \le \omega_0 < 1/8$. The function a_ρ is increasing, and its value at ρ_c is exactly one:

$$a_{\rho_c} = \frac{2\pi/(1+2\pi)}{2\pi/(1+2\pi)} = 1.$$

Thus $a_\rho \ge 1$. Formula (1) and $C_0 > 0$ give

$$\sqrt{K_0(\rho)} > \frac{\sqrt{a_\rho}}{\eta_\rho} > 8,$$

so $K_0(\rho) > 64$.

On the other hand, $\rho < 7/8$, and therefore

$$z_\rho < 8\pi < \frac{176}{7} < 26.$$

Consequently

$$k_{11}(\rho)^2 < 26^2 + 132 = 808 < 64^2,$$

which proves (2).

The small-hole wall H_0 is eligible only when $\rho < \omega_0$. But

$$\rho_c > \frac{1}{8} > \omega_0,$$

because $\pi < 22/7 < 7/2$ implies $1 + 2\pi < 8$. Also $\pi > 3$ gives the exact scope chain

$$\rho_c < \frac{1}{7} < \frac{39}{50} < \frac{5}{6}.$$

For $\rho < 5/6$, the thin-seam wall is not eligible either, so the piecewise upper-owner formula gives $U = K_0 > 64$. For $5/6 \le \rho < 7/8$, the seam branch owns the equality face and

$$U(\rho) = \frac{54}{(1-\rho)^2} \ge \frac{54}{(1/6)^2} = 1944 > 64.$$

Combining this with $k_{11} < 64$ proves (2). Finally $39/50 < 5/6$, so the final surviving interval lies wholly in the K_0 branch. □

2 Symbolic optical identities

The optical ledger evaluates the same algebraic statements at several sample values. The following identities replace those repetitions.

For $a/\pi = N + t$, with $N \ge 1$ and $0 < t \le 1$, the exact continuous-minus-product deficit is

$$\frac{D(a)}{\pi^2} = \frac{N^2}{2} + N\left(t^2 + \frac{1}{6}\right) + \frac{2t^3}{3}. \tag{3}$$

If $C_D = 1382/3125$, direct subtraction gives

$$\frac{D(a)}{\pi^2} - C_D(N+t)^2 = \left(\frac{1}{2} - C_D\right) N^2 + N\left(t^2 - 2C_Dt + \frac{1}{6}\right) + \frac{2t^3}{3} - C_Dt^2. \tag{4}$$

The coefficient of N^2 is positive. The difference between the right sides at $N+1$ and N is

$$2\left(\frac{1}{2} - C_D\right) N + \left(\frac{1}{2} - C_D\right) + t^2 - 2C_Dt + \frac{1}{6}.$$

The quadratic $t^2 - 2C_Dt + 1/6$ has minimum $1/6 - C_D^2$ at $t = C_D$. Consequently, for $N \geq 1$, the displayed difference is at least

$$3\left(\frac{1}{2} - C_D\right) + \frac{1}{6} - C_D^2 = \frac{4229603}{29296875} > 0.$$

Therefore the deficit increases in N, and it is enough to check $N = 1$. At $N = 1$, differentiation gives

$$\frac{d}{dt}\left(\frac{D(a)}{\pi^2} - C_D(1+t)^2\right) = 2(t - C_D)(t+1).$$

Thus the unique minimum on $0 < t \leq 1$ occurs at $t = C_D$, where the exact value is

$$\frac{1822532}{91552734375} > 0. \tag{5}$$

This proves the product screen simultaneously for every N, t.

Likewise, the twelve repeated angular block checks are all instances of

$$\sum_{\ell=0}^{m-1} (2\ell + 1) = m^2 \qquad (m \in \mathbb{N}_0), \tag{6}$$

proved by subtracting consecutive squares. At a radial equality wall the strict convention excludes the newly proposed layer, so no boundary sample is added. These symbolic statements cover every repeated optical row; the two exact endpoint reserves remain those printed in the main optical lemma.

3 The two optical screens with all scaling shown

For completeness, we now print the algebra that was compressed in the main optical lemma. Set

$$\varepsilon_0 = \frac{11}{50}, \qquad c = \frac{1126}{625}, \qquad q = \frac{106}{333}, \qquad C_D = \frac{1382}{3125}.$$

The inequalities

$$\pi > \frac{333}{106}, \qquad \pi < \frac{22}{7}$$

give $q > 1/\pi$, and $0 < \varepsilon \leq \varepsilon_0 < 1/2$.

Lemma 2 (Complete optical screen arithmetic)**.** *Let $a = \varepsilon K$.*

1. *If $\pi < a \leq c/\varepsilon$, then*

$$C_D > \frac{2cq}{3} + \frac{\varepsilon}{4} + \varepsilon^2 q^2. \tag{L}$$

2. If $a \geq c/\varepsilon$, define

$$L(\varepsilon) := \frac{(1-\varepsilon)^2}{4} - \frac{(22/7)(1-\varepsilon)\varepsilon}{4c},$$
$$A(\varepsilon) := q\varepsilon + \frac{\varepsilon^2}{4c} + \frac{q\varepsilon^3}{4c} + \frac{\varepsilon^4}{16c^2}.$$

Then $L(\varepsilon) > A(\varepsilon)$.

3. The low and high domains are both closed at $a = c/\varepsilon$, so their union covers every $a > \pi$.

Proof. For the low screen, $D(a) > C_D a^2$ was proved above. The sufficient product inequality obtained in the main optical argument is

$$D(a) \geq \varepsilon\left(\frac{2a^3}{3\pi} + \frac{a^2}{4}\right) + \frac{\varepsilon^2 a}{\pi}. \tag{1}$$

After division by a^2, the three terms on the right are bounded by

$$\frac{2\varepsilon a}{3\pi} \leq \frac{2cq}{3}, \qquad \frac{\varepsilon}{4} \leq \frac{\varepsilon_0}{4}, \qquad \frac{\varepsilon^2}{\pi a} < \frac{\varepsilon^2}{\pi^2} < \varepsilon_0^2 q^2.$$

Exact subtraction gives

$$C_D - \left(\frac{2cq}{3} + \frac{11}{200} + \left(\frac{11}{50}\right)^2 q^2\right) = \frac{39569}{2772225000} > 0. \tag{7}$$

This proves (L). For $0 \leq a \leq \pi$, the strict radial count is zero, including $a = \pi$, so no low-frequency gap remains.

For the high screen, the exact radial deficit and angular ceiling error from the main analytic argument are

$$\frac{D_{\text{rad}}}{a^2} \geq \frac{(1-\varepsilon)^2}{4} - \frac{\pi(1-\varepsilon)}{4a},$$
$$\frac{E}{a^2} = \frac{\varepsilon}{\pi} + \frac{\varepsilon^2}{4\pi a} + \frac{\varepsilon}{4a} + \frac{\varepsilon^2}{16a^2}.$$

Since $a \geq c/\varepsilon$, one has $1/a \leq \varepsilon/c$. Using the two rational bounds for π therefore gives

$$\frac{D_{\text{rad}}}{a^2} \geq L(\varepsilon), \qquad \frac{E}{a^2} < A(\varepsilon).$$

Moreover,

$$L'(\varepsilon) = -\frac{1-\varepsilon}{2} - \frac{22/7}{4c}(1-2\varepsilon) < 0 \quad (0 < \varepsilon < 1/2),$$

whereas $A'(\varepsilon) > 0$ coefficientwise. Hence

$$L(\varepsilon) - A(\varepsilon) \geq L(\varepsilon_0) - A(\varepsilon_0).$$

The endpoint subtraction is

$$\begin{aligned} L(\varepsilon_0) - A(\varepsilon_0) &= \frac{1}{4}\left(\frac{39}{50}\right)^2 - \frac{22}{7}\frac{(39/50)(11/50)}{4(1126/625)} \\ &\quad - \left[\frac{11}{50}q + \frac{(11/50)^2}{4(1126/625)} + \frac{(11/50)^3 q}{4(1126/625)} + \frac{(11/50)^4}{16(1126/625)^2}\right] \\ &= \frac{14817541}{472867032960000} > 0. \end{aligned} \tag{8}$$

Thus the radial deficit strictly exceeds the angular error.

Finally,

$$\frac{c}{\varepsilon} \geq \frac{c}{\varepsilon_0} = \frac{2252}{275} > \frac{22}{7} > \pi, \qquad \frac{2252}{275} - \frac{22}{7} = \frac{9714}{1925} > 0.$$

Therefore the common face lies in the low domain as well as the high domain, and

$$\{a : \pi < a \leq c/\varepsilon\} \cup \{a : a \geq c/\varepsilon\} = \{a : a > \pi\},$$

and equality $a = c/\varepsilon$ belongs to both pieces. Together with the zero-count interval $0 \leq a \leq \pi$, this covers every $a \geq 0$. At $K = a = 0$, both the spectral count and the Weyl term vanish. □

Ledger rows replaced. Lemma 1 replaces the eight executable `verify_u_and_k11` scope checks. Equations (3)–(6) replace the repeated product samples and odd-block identities. These are rows of `verify_optical_constants`. Lemma 2 prints the remaining scaling, monotonicity, common-face, and exact-reserve implications. Thus all fifty optical rows are consequences of displayed identities rather than executable predicates.